

PIACENZA SAN DAMIANO, Italy

WMO: 160840

Lat: 44.913N

Lon: 9.723E

Elev: 456

StdP: 14.46

Time Zone: 1.00 (EUC)

Period: 94-16

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 21.4	(c) 24.8	(d) 10.9	(e) 9.7	(f) 37.4	(g) 16.2	(h) 12.7	(i) 37.0	(j) 18.6	(k) 46.0	(l) 16.2	(m) 44.5	(n) 4.8	(o) 250	(p) 0.429

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.7	92.2	73.4	89.7	72.7	87.4	71.3	77.5	86.0	75.4	84.1	73.6	82.9	6.0	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
75.1	134.2	82.3	72.9	124.1	79.8	71.0	116.4	77.8	41.2	86.2	39.1	84.1	37.4	82.4	84.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
17.6	14.3	12.0	DB	17.7	95.7	4.3	3.6	14.6	98.3	12.1	100.4	9.6	102.4	6.5	105.0	
			WB	17.2	79.1	4.2	2.0	14.2	80.5	11.7	81.7	9.3	82.8	6.2	84.3	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	55.8	37.4	39.6	47.6	54.1	63.2	70.6	74.7	65.6	56.4	46.9	38.2		(6)
	DBStd	14.41	4.60	6.44	5.78	5.25	4.84	5.34	4.59	5.10	4.87	6.23	5.77	5.59	(7)
	HDD50	1323	391	292	113	22	0	0	0	0	17	123	365		(8)
	HDD65	4237	856	711	538	329	92	13	1	2	50	271	544	830	(9)
	CDD50	3456	0	1	39	144	410	617	765	468	216	30	1		(10)
	CDD65	893	0	0	0	0	37	180	301	302	68	5	0	0	(11)
	CDH74	9051	0	0	6	19	406	1797	3142	3134	513	34	0	0	(12)
	CDH80	3364	0	0	0	2	69	614	1249	1336	90	4	0	0	(13)
(14) Wind	WSAvg	4.3	3.1	4.1	5.0	5.2	5.1	4.8	5.0	4.7	4.5	3.6	3.3	2.7	(14)
(15) Precipitation	PrecAvg	34.6	2.3	2.2	2.5	3.5	2.8	3.0	1.3	2.3	3.4	4.7	4.8	2.4	(15)
	PrecMax	51.8	6.5	7.2	7.7	7.5	5.8	9.3	4.9	5.1	10.9	9.8	9.9	7.9	(16)
	PrecMin	23.5	0.1	0.0	0.2	0.1	0.2	0.7	0.2	0.3	0.4	1.1	1.2	0.0	(17)
	PrecStd	8.3	1.7	2.0	1.9	1.7	1.5	2.4	1.1	1.5	2.4	2.4	2.6	1.9	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.8	63.2	73.3	76.8	85.8	93.3	93.5	97.2	86.0	79.1	62.8	55.2	(19)
		MCWB	44.8	50.1	55.2	58.6	67.1	71.0	76.3	73.2	70.1	63.4	53.7	48.1	(20)
	2%	DB	50.2	57.1	67.2	73.0	81.5	89.5	91.4	93.4	82.6	72.0	59.1	50.4	(21)
		MCWB	43.4	46.4	53.5	57.4	64.5	72.1	74.3	73.4	67.4	63.7	55.2	47.4	(22)
	5%	DB	47.0	53.4	64.1	69.6	78.6	86.2	89.4	90.1	80.0	69.4	57.1	48.3	(23)
		MCWB	43.2	45.7	51.7	56.7	63.6	71.0	73.0	72.5	66.0	62.6	54.3	46.2	(24)
	10%	DB	44.6	49.6	60.5	66.4	75.4	83.7	86.8	87.5	76.8	66.1	55.2	46.3	(25)
		MCWB	42.1	44.1	50.3	55.2	63.0	69.4	71.5	71.1	64.6	60.5	53.2	44.7	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	49.4	52.4	59.0	62.1	71.7	78.2	82.0	78.6	73.7	68.6	58.9	51.9	(27)
		MCDB	50.9	59.2	68.4	71.4	80.3	84.8	89.9	88.5	81.5	71.6	59.7	52.8	(28)
	2%	WB	46.5	49.3	55.7	60.1	68.7	74.7	78.5	76.5	71.1	66.1	56.8	49.0	(29)
		MCDB	47.7	55.1	63.2	67.6	75.5	82.9	86.4	86.3	77.9	69.0	58.1	49.6	(30)
	5%	WB	44.4	46.8	53.4	58.3	66.5	72.9	76.2	74.7	69.1	64.3	55.2	46.8	(31)
		MCDB	45.6	50.0	60.9	65.8	74.3	81.6	83.9	84.8	75.5	67.9	56.3	47.5	(32)
	10%	WB	42.4	45.1	51.5	56.8	64.6	71.4	73.8	73.3	67.3	62.3	53.5	44.8	(33)
		MCDB	43.8	47.7	58.3	64.0	72.4	80.2	82.5	83.5	72.8	65.4	54.6	45.5	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	11.9	15.6	20.1	19.4	20.6	21.0	21.7	21.1	19.1	14.1	10.5	10.6	(35)
		MCDBR	17.2	23.3	26.6	25.7	25.2	24.5	23.3	25.2	23.2	18.2	13.7	14.0	(36)
		MCWBR	12.1	14.6	14.5	13.0	11.9	10.5	9.6	9.4	10.0	10.5	9.7	10.2	(37)
	5% WB	MCDBR	14.1	18.6	22.8	21.5	21.9	21.5	20.9	22.2	19.1	15.4	9.7	10.2	(38)
		MCWBR	11.4	13.1	14.2	12.4	12.7	11.3	10.4	10.5	10.6	10.2	8.2	8.5	(39)
(40) Clear-Sky Solar Irradiance	taub		0.353	0.381	0.428	0.459	0.454	0.476	0.460	0.450	0.440	0.430	0.381	0.347	(40)
	taud		2.353	2.246	2.121	2.067	2.126	2.116	2.170	2.210	2.204	2.185	2.283	2.383	(41)
	Ebn at Noon		235	249	252	254	260	254	257	255	245	227	220	225	(42)
	Edh at Noon		25	33	43	49	48	49	46	42	39	35	27	23	(43)
(44) All-Sky Solar Radiation	RadAvg		465	715	1128	1451	1799	2072	2131	1814	1349	797	475	393	(44)
	RadStd		65	110	156	160	162	117	121	115	89	119	73	55	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
Regional (36 neighbors)	+0.59	N/A	N/A	N/A	+1.04	N/A	N/A	N/A	N/A	+113	+14

Nomenclature: See separate page